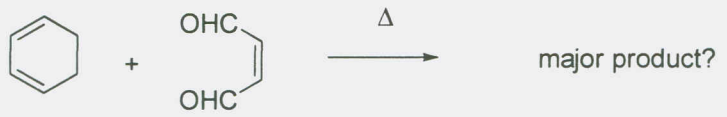
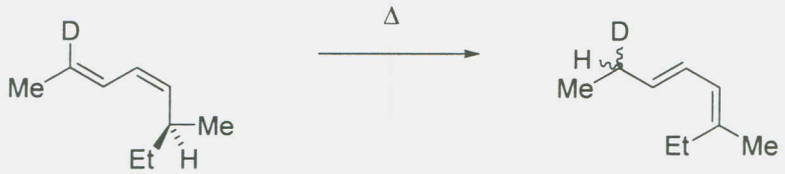
Organic Cume
9-18-2009

1. For the reaction shown below, (a) give the major product including stereochemistry and (b) state why the alkene has carbaldehyde substitutents and show appropriate diagrams that support your conclusion would changing the carbaldehyde substituents to methoxy substituents result in a slower or faster reaction? (b) State and show with diagrams clearly how you came to your conclusion using orbital symmetry theory.



2. For the below reaction, give the expected stereochemistry of the product and show how you came to your conclusion by using appropriate orbital symmetry theory.

3. Predict the following reaction stereochemistry and show how using orbital symmetry theory this can be predicted.



4. For the following nucleophiles, I^-, Br^-, Cl^-, rank them in terms of increasing nucleophilic strength in protic solvents. State the reasons for your ranking of each.

Would you expect this same trend in aprotic solvents? Why?